Hrishikesh Naveenam

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EDUCATION

The University of Texas at Dallas

B.S. Computer Science, Minor in Mathematics | May 2028 | GPA: 3.72

- Coursework: Data Structures and Algorithms, Object-Oriented Programming, Discrete Mathematics, Computer Architecture, Linear Algebra, Probability and Statistics.

RESEARCH EXPERIENCE

Undergraduate Research Assistant, UTD URAP

May 2026 - Present | Advised by Prof. Wei Yang

- Curated 200+ documented root causes from issues, commits, and pull requests across open-source ML and inference repositories.
- Developed a taxonomy covering floating-point reduction order, kernel selection, batching and scheduling, and sampling and seed handling.
- Built Python harnesses that isolate and replay divergent inference runs.

Student Researcher, UT Dallas

February - May 2025 | CelestAI / Galaxy10 dataset

- Trained a DCGAN in PyTorch, reaching 0.85 ROC-AUC and 0.72 recall on Galaxy10.
- Fine-tuned StyleGAN to reach 0.81 ROC-AUC on rare celestial events.
- Improved rare-event detection performance by 15% over baseline with GAN-based data augmentation.

Research Assistant, University of Missouri

September 2023 - January 2024 | Distributed scientific workflows

- Reached 84% workflow-placement accuracy using K-Means clustering over CPU, RAM, and storage telemetry.
- Built a repeatable Docker and AWS EC2 testbed with multiple Linux VMs, using the Elbow Method to tune cluster size.

OPEN SOURCE

OpenWorker - Contributor

Python / FastAPI / REST | PR #185

- Contributed four NVIDIA NIM models through an OpenAI-compatible provider interface.
- First fixed an authentication gap that accepted invalid API keys, using an inference-based verification probe and dedicated unit tests.

github.com/andrewyng/openworker/pull/185

HONORS AND AWARDS

2nd Place, PNC Track - HackUTD 2025 (350+ participants) | November 2025

Dean's List, UT Dallas | Fall 2024

PROJECTS

Holler

Python / FastAPI / TypeScript / SQLite / Twilio / Composio

- A persistent agent runtime for WhatsApp, Telegram, and Twilio voice, with Gmail, Outlook, Drive, and Calendar tools.
- Reached 91% task completion on a 100-workflow evaluation, with 28% fewer LLM calls and 24% lower inference cost than a multi-agent baseline.
- Used single-use SQLite action records to prevent duplicate execution after a crash.

github.com/hrishinave/Holler

Live multiplayer poker

July 2026 | React / TypeScript / Node.js / WebSockets / Supabase / Vercel

- A multiplayer poker app with an authoritative WebSocket server, persistent accounts, and server-controlled chip balances.
- Load-tested 360 concurrent players across 60 tables at 1,871 actions per second, with 0.3 ms p95 server action-to-broadcast latency.
- Validated chip conservation and game rules across 3,000 fuzzed hands and 106 tests, including reconnects, restarts, and rate limiting.

poka-client.vercel.app

AutoPM

November 2025 | Next.js / TypeScript / LangGraph / Gemini / Auth0

- Eight LangGraph agents run a seven-step product workflow: research, user stories, prioritization, wireframes, and Jira tickets.
- Estimated to save 10+ hours of manual product work per cycle.
- Connected Jira, Gmail, and Gemini through Zod-validated structured outputs, replacing 5+ manual handoffs.

NetGainNBA

February - May 2025 | React / Flask / Python / XGBoost / scikit-learn

- NBA playoff predictions and a general-manager toolkit with bracket simulation, salary-cap tracking, and a trade engine.
- Reached 88% playoff-prediction accuracy on 2020-2024 data using a seven-model ensemble across 89 features.
- Built an NBA Stats API pipeline with nine player archetypes and live win-probability tracking.

net-gain-nba.vercel.app

EduTube

September 2025 | React / TypeScript / Fastify / Google Cloud / TwelveLabs / Gemini

- A lecture companion with semantic video search and generated study materials, built with TwelveLabs and Gemini.
- Processed 100+ educational videos.
- Built a Fastify backend with Google Cloud Storage and webhooks for real-time video processing.

LEADERSHIP AND INVOLVEMENT

Lead Developer - AI51 Innovation Labs

August 2025 - Present | Research Mate

- Building a research paper assistant that answers questions with citations to specific source pages.
- Built a LangGraph retrieval pipeline using a hybrid ColQwen2 index across Qdrant and FAISS; answers cite the retrieved source pages.
- Made page ingestion idempotent with Postgres state tracking and deterministic vector IDs.
- Implemented background page indexing and enforced document ownership through Supabase authentication.

Product Manager - UT Dallas AI Mentorship Program

August 2025 - Present | TuneTrend

- Leading a team of 4-5 engineers building a genre-aware music popularity predictor.
- Designed an ML pipeline extracting 140+ audio features, with artist-level data splits for evaluation.
- Shipped a platform serving six genre models, with audio hook detection, 3D visualizations, and nearest-neighbor track matching.

Officer - Artificial Intelligence Society, UT Dallas

May 2025 - Present | HackAI & technical workshops

- Helping organize HackAI for hundreds of participants, covering event logistics, corporate sponsorships, and technical challenges.
- Facilitating technical workshops and student projects in AI.

TECHNICAL SKILLS

Languages

Python, Java, C++, JavaScript, TypeScript, SQL, Swift, HTML/CSS

Machine learning & data

PyTorch, scikit-learn, XGBoost, pandas, NumPy, LangGraph, RAG, Qdrant, FAISS, DCGAN, StyleGAN, librosa

Web & backend

React, Next.js, Node.js, Flask, FastAPI, Fastify, PostgreSQL, SQLite, Supabase, WebSockets, Auth0, Zod

Infrastructure & tools

AWS, Google Cloud Storage, Docker, Kubernetes, Linux, Git, CI/CD, Vercel, Twilio, Composio